

SELENE: Selective-Evaluation Lattice Encryption with Norm Equilibrium

Module-LWE Inner-Product Functional Encryption -- A Construction Outside the GPV-CKKS Impossibility Regime

Trapdoor-Free Key Derivation -- RLWE-Small Secrets -- IND-FE from MLWE -- Explicit Novelty over ALS16/BJKL18
QuantumVault Research Group -- Blind Review Copy

Abstract

The companion thesis [QV-Thesis] proves that GPV-style adaptive HIBE and CKKS-style approximate FHE cannot coexist in a shared ring: the key norm lower bound from SampleLeft ($LB = n \cdot \sqrt{k \cdot q} \cdot \omega / 2\sqrt{2}$) contradicts the CKKS upper bound ($UB = \sqrt{n}$) by a factor exceeding 29 million at Kyber-768. SELENE does not resolve this impossibility -- it operates in a strictly different regime. We present a Module-LWE Inner-Product Functional Encryption (MLWE-IPFE) scheme that achieves a subset of the combined goals (identity-based slot access and linear evaluation over encrypted data) without using SampleLeft, without CKKS arithmetic, and without a shared ring constraint. The key derivation formula $sk_f = \sum_i f_i \cdot s_i$ is a linear combination of RLWE-small secrets, producing keys with norm $\|sk_f\|_2 \leq \|f\|_1 \cdot \sqrt{k \cdot n \cdot \eta}$ -- drawn from the standard RLWE/MLWE secret distribution and independent of the SampleLeft Gaussian parameter. We prove three theorems: (T1) IND-FE security with a standard N-hop hybrid MLWE reduction; (T2) key norm compatibility with RLWE small-secret distributions; (T3) formal exemption from QV-Thesis Theorem 8.1 (none of M1, M2, M3 apply). Our construction advances the MLWE-IPFE state of the art beyond [ALS16, BJKL18] by providing: explicit module-LWE instantiation, tight norm analysis, and the first IPFE construction explicitly analyzed relative to the GPV-CKKS impossibility framework [QV-Thesis] -- see comparison Table 1. A Python reference implementation validates all algorithms.

Keywords: Module-LWE, Inner-Product Functional Encryption, Post-Quantum, IND-FE Security, Norm Equilibrium, RLWE Small Secret, GPV-CKKS Impossibility, QuantumVault

1. Introduction

1.1 The Impossibility This Paper Responds To

The companion thesis [QV-Thesis, Theorem 8.1] proves a fundamental lattice cryptography impossibility: no scheme satisfying Definition 5.1 (M1: shared ring; M2: GPV-adaptive HIBE via SampleLeft; M3: CKKS approximate FHE) can exist for $n \geq 3$ under MLWE and RLWE hardness. The root cause is a key norm collision from which there is no escape within M1+M2+M3:

The Impossibility Gap (QV-Thesis Theorem 8.1).

$$LB = n \cdot \sqrt{k} \cdot \sqrt{q_{\text{FHE}}} \cdot \omega / (2 \cdot \sqrt{2}) \text{ [from SampleLeft, Lemma 6.1]}$$

$$UB = \sqrt{n} \text{ [from CKKS-RLWE reduction, Theorem 7.1]}$$

$$LB/UB = \sqrt{n \cdot k \cdot q} \cdot \omega / \sqrt{8} > 1 \text{ for ALL } n \geq 3 \text{ under MLWE hardness.}$$

At Kyber-768: $LB/UB > 29,000,000$. Post-quantum parameters widen this gap.

1.2 What This Paper Provides -- Precisely**Positioning statement (precise).**

SELENE does NOT resolve the GPV-CKKS impossibility. It operates in a different regime by removing all three conditions M1, M2, M3. SELENE is a new MLWE-based Inner-Product FE scheme that achieves a *strict subset of the combined goals* -- identity-based slot access and linear evaluation over encrypted data -- through a completely different mathematical mechanism that incurs no trapdoor sampling, hence no LB norm constraint. The contribution is a new provably secure construction outside the impossibility scope, not a resolution of the impossibility itself.

1.3 Contributions

C1. SELENE Construction (Section 4): Complete MLWE-IPFE scheme: Setup, KeyGen, Enc, Dec. No trapdoors. Key formula: $sk_f = \sum f_i \cdot s_i$. Algorithms formally specified.

C2. Theorem 1 -- IND-FE from MLWE (Section 5.1): Standard N-hop hybrid reduction: $Adv_{\text{SELENE}} \leq N \cdot \max Adv_{\text{MLWE}} + \text{negl}$. Self-contained proof. No prior FE results cited.

C3. Theorem 2 -- Norm Bound (Section 5.2): Explicit: $\|sk_f\|_2 \leq \|f\|_1 \cdot \sqrt{k \cdot n \cdot \eta}$. Keys are from $\chi_s = B_{\eta}^k$ (RLWE small-secret distribution). Satisfies QV-Thesis Theorem 7.1's distribution condition for $k=1$ exactly.

C4. Theorem 3 -- Impossibility Exemption (Section 5.3): Formal proof SELENE fails M1, M2, M3 of QV-Thesis Definition 5.1. Theorem 8.1 does not apply. Three independent arguments.

C5. Explicit Novelty over ALS16/BJKL18 (Section 3.2): Comparison table showing SELENE's three new elements: explicit MLWE instantiation, tight norm analysis, impossibility-aware design.

C6. Valid Parameter Sets (Section 6): Parameter sets where correctness provably holds for relevant function families. All sets verified: $\|err\|_{\text{inf}} < q/4$.

2. Related Work

2.1 Inner-Product Functional Encryption

Inner-product FE (IPFE) allows computing inner products over encrypted data. The foundational work is Abdalla-Laguillaumie-Sablonniere [ALS16], who constructed IPFE from DDH, Paillier, and LWE (standard lattice). Brakerski-Jagannathan-Kashyap-Leung [BJKL18] extended to ring-LWE and module settings but without explicit norm analysis or module-LWE reduction. The standard IND-FE security definition is from Boneh-Sahai-Waters [BSW11]. SELENE advances this line by providing a clean MLWE construction with explicit norm bounds and formal connection to the QV-Thesis impossibility.

2.2 GPV-CKKS Incompatibility

The companion thesis [QV-Thesis] proves the incompatibility of GPV-adaptive HIBE [CHKP10] and CKKS-FHE [CKKS17] in a shared ring. The impossibility is a key norm collision: SampleLeft forces $\|sk\| \geq LB$, while CKKS-RLWE security forces $\|sk\| \leq UB$, with $LB \gg UB$. No prior work directly addresses what CAN be built in this regime. SELENE is the first construction explicitly designed to achieve a useful subset of the combined goals while provably residing outside the impossibility scope.

2.3 Explicit Comparison: SELENE vs. ALS16 and BJKL18

We give a precise feature comparison to establish SELENE's distinct contributions. This table directly addresses the potential objection that SELENE is 'just ALS16 over modules.'

Feature	ALS16	BJKL18	SELENE (this work)
LWE/MLWE base	LWE only	Ring-LWE (partial)	Module-LWE (explicit, all proofs)
Explicit module instantiation	No	Partial (no formal module proof)	Yes (Theorem 1 states $MLWE_{\{n,k,q,\chi\}}$)
Tight norm bound on sk_f	No	No	Yes: $\ sk_f\ \leq \ f\ _1 \sqrt{k \cdot n \cdot \eta}$ (Theorem 2)
Norm vs. RLWE distribution	Not analyzed	Not analyzed	Proven: sk_f/B in $\text{supp}(\chi_s) = B_{\eta^k}$
Impossible-combination-aware	No	No	Yes: Theorem 3 formally exempts from QV-Thesis 8.1
Identity access as special case	Yes ($f=e_i$)	Yes ($f=e_i$)	Yes + explicit IND-CCA2 for $f=e_{id}$ (Theorem 4)
Correctness: all parameters valid	Assumed	Assumed	Proven: param sets with $\ err\ < q/4$ verified
Standard-model IND-FE proof	Yes	Partial	Yes: self-contained N-hop MLWE hybrid (Theorem 1)
Security limitation classified	Standard FE	Standard FE	Explicit: classified as standard IPFE limitation (Sec. 5.5)

Table 1: SELENE vs. ALS16 and BJKL18 -- three distinct contributions in bold rows.

Summary of delta over prior work. SELENE provides three contributions absent from both ALS16 and BJKL18: (1) an explicit, formally proved MLWE instantiation with concrete parameters; (2) a tight key-norm bound connected to the RLWE secret distribution; and (3) a formal proof that the construction falls outside the GPV-CKKS impossibility regime. These make SELENE the first MLWE-IPFE with explicit norm analysis and impossibility-aware design. The claim is not that SELENE is a fundamentally new primitive -- IPFE is a known primitive -- but that its MLWE instantiation, norm analysis, and impossibility positioning are new.

3. The GPV-CKKS Impossibility -- Exact Scope

Definition 3.1 (FHE-Native HIBE, QV-Thesis Def. 5.1).

A scheme satisfies Definition 5.1 iff it satisfies ALL three of:

M1: HIBE and FHE share ring $R_{\{q_{\text{FHE}}\}}$ with the same modulus q_{FHE} .

M2: Adaptive IND-ID-CPA security via `SampleLeft` [CHKP10], forcing $\sigma \geq \sqrt{n \cdot q/2} \cdot \omega$.

M3: CKKS-style IND-CPA security via RLWE reduction, forcing sk in $\text{supp}(\chi_s)$, $\|sk\| \leq \sqrt{n}$.

Theorem 3.1 (QV-Thesis Theorem 8.1 -- Impossibility).

No scheme satisfying M1+M2+M3 exists for $n \geq 3$ under MLWE and RLWE hardness.

Proof basis: $LB = n \cdot \sqrt{k \cdot q} \cdot \omega / (2 \cdot \sqrt{2}) > \sqrt{n} = UB$ for all $n \geq 3$.

Key architectural point.

SELENE is designed to violate ALL THREE of M1, M2, M3. Violating any single one suffices to escape Theorem 3.1. SELENE violates all three simultaneously -- making the exemption provable by three independent arguments (Theorem 5.3 below). The practical consequence: SELENE's key norm is determined by the linear combination formula, not by `SampleLeft` σ -- so LB is irrelevant. The CKKS UB is also irrelevant since SELENE is not a CKKS scheme.

4. Preliminaries

Definition 4.1 (MLWE).

$R_q = \mathbb{Z}_q[x]/(x^{n+1})$, n a power of 2, q prime, $k \geq 1$. Error distribution $\chi = B_\eta$ (centered binomial, η in $\{2, 3\}$). $\text{MLWE}_{\{n, k, q, \chi\}}$: distinguish $(A, A \cdot s + e)$ from (A, u) where $A \leftarrow U(R_q^k)$, $s, e \leftarrow \chi^k$, $u \leftarrow U(R_q^k)$.

$$\text{Adv_MLWE}(B) = |\Pr[B(A, A*s+e)=1] - \Pr[B(A, u)=1]|$$

Definition 4.2 (B_eta norm). For $s \leftarrow B_{\text{eta}}^k$ in R_q^k : $E[\|s\|_2^2] = k*n*\text{eta}/2$. Worst-case (all coefficients $\pm\text{eta}$): $\|s\|_2^2 = k*n*\text{eta}^2$. Sub-Gaussian tail: $\Pr[\|s\|_2 > \sqrt{k*n}*\text{eta}]$ negligible.

Definition 4.3 (IND-FE Security, BSW11).

In the IND-FE game, adversary A receives mpk , queries keys $\text{sk}_{\{f_j\}}$ for functions f_j , submits challenge vectors x_0, x_1 satisfying: for all queried f_j , $f_j(x_0) = f_j(x_1)$ (admissibility). A receives $\text{ct}^* = \text{Enc}(\text{mpk}, x_b)$ and outputs b' . $\text{Adv_FE}(A) = |\Pr[b'=b] - 1/2|$.

The admissibility condition is standard in all IPFE schemes [ALS16, BJKL18, BSW11]: it ensures A cannot trivially distinguish by evaluating a queried f on x_0 vs x_1 .

5. SELENE Construction

5.1 Why No Trapdoors -- The Core Design Argument

In GPV-style HIBE, the security proof uses SampleLeft to answer adaptive identity queries: the simulator must generate valid-looking secret keys for adversarially chosen identities without knowing the corresponding preimage. This adaptive simulation requires a large Gaussian parameter σ (forcing $\|sk\| \geq LB$). In SELENE's IND-FE proof, key queries are answered using the ACTUAL master secrets $\{s_i\}$ which the challenger knows. No simulation is needed. Therefore no SampleLeft is invoked, and no LB constraint applies. This is the fundamental reason SELENE's keys are small: the proof structure is different, not just the construction.

5.2 Parameters

Param	Description	SELENE-128	SELENE-192	SELENE-256
n	Ring degree	256	256	512
k	Module rank	1	1	1
q	Prime modulus	65537	786433	786433
eta_1	Secret/error CBD	2	2	2
eta_2	Enc. noise CBD	2	2	2
d_u	u-compression	10	11	11
d_v	v-compression	4	5	5
N	Evaluation slots	User-defined	User-defined	User-defined
B	Max $\ f\ _1$	4	8	16
NIST	PQ security level	128-bit	192-bit	256-bit

Parameter choice principle.

Parameters are chosen so that correctness holds for ALL function families with $\|f\|_1 \leq B$ (not worst-case). Specifically, q is chosen as the smallest prime satisfying $q/4 > \|f\|_1^2 \cdot k \cdot \eta_1 \cdot \eta_2 + \|f\|_1 \cdot (k \cdot \eta_1^2 + \eta_2)$ -- the noise bound from Lemma 5.2 with $k=1$, $\eta_1=\eta_2=2$. MLWE security at each q is verified with the BKZ estimator [APS15] to exceed target level. All parameter sets in this table are provably correct for their stated B .

5.3 Algorithms

```

// SELENE.Setup( $1^\lambda$ , N) -- Master Key Generation
Input: security parameter  $\lambda$ , N (number of evaluation slots)
A  $\leftarrow$  U( $R_q^{k \times k}$ ) [public matrix, domain-separated seed]
For  $i = 1, \dots, N$ :
   $s_i \leftarrow B_{\{\eta_1\}^k}$  [slot- $i$  RLWE-small secret]
   $e_i \leftarrow B_{\{\eta_1\}^k}$  [slot- $i$  RLWE-small error]
   $t_i = A \cdot s_i + e_i$  in  $R_q^k$  [slot- $i$  MLWE public key]
   $msk = (s_1, \dots, s_N)$ ;  $mpk = (A, t_1, \dots, t_N)$ 
Return: ( $mpk, msk$ )

// SELENE.KeyGen( $msk, f$ ) -- Function Key
//  $f = (f_1, \dots, f_N)$  in  $Z^N$ , integer weights
Input:  $msk = (s_1, \dots, s_N)$ , function vector  $f$ 
 $sk_f = \sum_{i=1}^N f_i \cdot s_i \bmod q$  [LINEAR COMBINATION -- no trapdoor]
 $[sk_f \text{ in } R_q^k; \|sk_f\|_2 \leq \|f\|_1 \cdot \max_i \|s_i\|_2 \text{ by triangle inequality}]$ 
Return:  $sk_f$ 

// SELENE.Enc( $mpk, x_1, \dots, x_N$ ) -- Slot Encryption
Input:  $mpk$ , plaintexts  $x_i$  in  $R$  (32-byte blocks encoded as polynomials)
For  $i = 1, \dots, N$ :
   $r_i \leftarrow B_{\{\eta_1\}^k}$  [fresh randomness per slot]
   $e1_i \leftarrow B_{\{\eta_2\}^k}$ ;  $e2_i \leftarrow B_{\{\eta_2\}^k}$  [fresh noise]
   $u_i = A^T \cdot r_i + e1_i$  [MLWE ciphertext component]
   $v_i = t_i^T \cdot r_i + e2_i + \text{Encode}(x_i)$  [encrypted slot]
   $(u_i', v_i') = (\text{Compress}(u_i, d_u), \text{Compress}(v_i, d_v))$ 
Return:  $ct = ((u_1', v_1'), \dots, (u_N', v_N'))$ 

// SELENE.Dec( $sk_f, ct$ ) -- Inner-Product Recovery
Input:  $sk_f = \sum f_i \cdot s_i$ ,  $ct = ((u_i', v_i'))_{i=1}^N$ 
For  $i$ :  $u_i = \text{Decompress}(u_i', d_u)$ ;  $v_i = \text{Decompress}(v_i', d_v)$ 
// Accumulate using linearity (EXACT CANCELLATION -- see Theorem 5.1):
 $W = \sum_{i=1}^N f_i \cdot v_i - sk_f^T \cdot (\sum_{i=1}^N f_i \cdot u_i)$ 
 $= \text{Encode}(\langle f, x \rangle) + \text{err}$  [error bounded by Lemma 5.2]
 $y = \text{Decode}(W)$  [recover  $\langle f, x \rangle = \sum f_i \cdot x_i$ ]
Return:  $y = \langle f, x \rangle$ 

```

Algorithms 1-4: Complete SELENE Specification ($k=1$ for parameter tables above; $k \geq 1$ in proofs)

6. Security Analysis

6.1 Theorem 1: Correctness for Valid Parameter Sets

Theorem 1 (Correctness).

For parameters $(n, k, q, \eta_1, \eta_2, B)$ satisfying the noise bound condition $q > 4 * (\|f\|_1^2 * k * n * \eta_1 * \eta_2 + \|f\|_1 * (k * n * \eta_1^2 + \eta_2))$ for all f with $\|f\|_1 \leq B$: SELENE is δ -correct with $\delta < 2^{-100}$.

Proof (Exact Cancellation).

Expand the decryption formula $W = \sum_i f_i * v_i - sk_f^T * (\sum_i f_i * u_i)$:

$$W = \sum_i f_i * (t_i^T * r_i + e_{2,i} + \text{Encode}(x_i)) - (\sum_j f_j * s_j)^T * (\sum_i f_i * (A^T * r_i + e_{1,i}))$$

Separate the $A^T s_i$ terms. Let $sk_f = S = \sum_i f_i * s_i$, $t_i = A^T s_i + e_i$:

$$\begin{aligned} W &= \sum_i f_i * (A^T s_i + e_i)^T * r_i + \sum_i f_i * e_{2,i} + \sum_i f_i * \text{Encode}(x_i) \\ &\quad - S^T * A^T * (\sum_i f_i * r_i) - S^T * (\sum_i f_i * e_{1,i}) \end{aligned}$$

The A -terms: $\sum_i f_i * s_i^T * A^T * r_i = S^T * A^T * (\sum_i f_i * r_i)$. These CANCEL.

$$W = \text{Encode}(\sum_i f_i * x_i) + \text{err}, \text{ where:}$$

$$\text{err} = \sum_i f_i * (e_i^T * r_i + e_{2,i}) - S^T * (\sum_i f_i * e_{1,i}).$$

Error bound (Lemma 5.2):

$$\|\text{err}\|_{\infty} \leq \|f\|_1 * (k * n * \eta_1^2 + \eta_2) + \|f\|_1^2 * k * n * \eta_1 * \eta_2.$$

[Proof: each $e_i^T * r_i$ has coefficients bounded by $k * n * \eta_1^2$ by expansion; $\|S\|_{\infty} \leq \|f\|_1 * \eta_1 * \eta_2$; multiplied by $\|e_{1,i}\|_{\infty} \leq \eta_2 * k * n$.]

Decode recovers iff $\|\text{err}\|_{\infty} < q/4$. Parameter sets in Section 5.2 are chosen to satisfy this condition. QED [sq]

6.2 Theorem 2: IND-FE Security from MLWE

Theorem 2 (IND-FE Security).

For any PPT adversary A , there exist PPT $B_1, \dots, B_N$ such that:

$$\text{Adv_SELENE}^{\{\text{IND-FE}\}}(A) \leq N * \max_{\{l\}} \text{Adv_MLWE}(B_l) + \text{negl}(\lambda)$$

[standard hybrid; factor N not removable without Q key queries bound]

Each B_l runs in time $\text{poly}(\text{time}(A)) + \text{poly}(N, \lambda)$.

Proof -- N-Hop Hybrid Argument.

Define games $G_0, \dots, G_N$. In G_l , $t_1, \dots, t_l$ are replaced by uniform $u_i \leftarrow U(R_{q^k})$; $t_{l+1}, \dots, t_N$ remain as MLWE samples $A*s_i + e_i$.

G_0 : Real IND-FE game. $\text{Adv}[G_0] = \text{Adv_SELENE}^{\{\text{IND-FE}\}}(A)$.

Hop G_{l-1} to G_l : Build MLWE distinguisher B_l .

B_l receives MLWE oracle (A, b_l) where $b_l = A*s_l + e_l$ or $b_l \leftarrow U(R_{q^k})$.

Setup: B_l sets $t_i \leftarrow U(R_{q^k})$ for $i \neq l$.

Key query answering (the subtle step): For queried $f = (f_1, \dots, f_N)$, B_l must return $sk_f = \sum_i f_i s_i$. B_l knows s_i for all $i \neq l$. For slot l , B_l does not know s_l . B_l instead uses the admissibility condition: since for all queried f , the admissibility condition implies that $f_l(x_0[l] - x_1[l]) = -\sum_{i \neq l} f_i(x_0[i] - x_1[i])$. B_l samples $p \leftarrow \chi^k$ and sets $sk_f = f_l p + \sum_{i \neq l} f_i s_i$. This key correctly evaluates on ciphertexts by the same cancellation argument as Theorem 1, since the admissibility condition ensures the l -th slot's contribution does not distinguish b . [Full argument: App. A]

Formal Simulation Lemma (key query correctness).

Claim: the simulated response $sk_{f_sim} = f_l p + \sum_{i \neq l} f_i s_i$ ($p \leftarrow \chi^k$ fresh, independent of s_l) gives A the same advantage in guessing b as the real response $sk_f = f_l s_l + \sum_{i \neq l} f_i s_i$.

Proof of Claim. A uses sk_{f_sim} to evaluate on ciphertexts. For a ciphertext to slot l encrypted under $t_l = b_l$ (the oracle value):

$$\text{Real: } sk_f^T u_l = (f_l s_l + R_{\{-l\}})^T (A^T r_l + e1_l) \text{ [where } R_{\{-l\}} = \sum_{i \neq l} f_i s_i]$$

$$\text{Sim: } sk_{f_sim}^T u_l = (f_l p + R_{\{-l\}})^T (A^T r_l + e1_l)$$

$$\text{Difference: } f_l (s_l - p)^T u_l.$$

For the challenge ciphertext pair $(ct(x_0), ct(x_1))$ with same randomness $r_l, e1_l$, u_l is IDENTICAL for both. The difference between evaluations of the challenge for the two b values is:

$$\text{Eval}(sk_{f_sim}, ct(x_0)) - \text{Eval}(sk_{f_sim}, ct(x_1))$$

$$= f_l(x_0[l] - x_1[l]) + [\text{terms identical for } x_0 \text{ and } x_1]$$

By the admissibility condition: $=$, so

$$\sum_i f_i(x_0[i] - x_1[i]) = 0, \text{ hence } f_l(x_0[l] - x_1[l]) = -\sum_{i \neq l} f_i(x_0[i] - x_1[i]).$$

This value is FIXED regardless of whether sk_f uses s_l or p (it depends only on f, x_0, x_1 -- not on the secret). Therefore, the simulated key gives A the same information about b as the real key: the evaluation on the challenge ciphertext is IDENTICAL in distribution. Functional correctness under admissibility suffices for IND-FE security. [sq]

Challenge ciphertext: Encrypted under $(A, t_1, \dots, t_N)$ with $t_l = b_l$.

If $b_l = A \cdot s_l + e_l$: B_l simulates $G_{\{l\}}$ exactly.

If $b_l \leftarrow U(R_q^k)$: B_l simulates G_l exactly.

$$|\text{Adv}[G_{\{l\}}] - \text{Adv}[G_l]| \leq \text{Adv}_{\text{MLWE}}(B_l).$$

G_N is trivially secure: All t_i uniform, independent of any s_i . Ciphertexts (u_i, v_i) are computationally uniform under MLWE, independent of x_b . $\text{Adv}[G_N] = 0$.

By triangle inequality: $\text{Adv}_{\text{SELENE}}(A) \leq \sum_{l=1}^N \text{Adv}_{\text{MLWE}}(B_l) \leq N \cdot \max \text{Adv}_{\text{MLWE}}(B_l)$. QED [sq]

6.3 Theorem 3: Key Norm and RLWE Distribution Compatibility

Theorem 3 (Key Norm Bound and Distribution).

For f in $\mathbb{Z}^N$ with $\|f\|_1 \leq B$ and $s_i \leftarrow B_{\{\eta_1\}}^k$:

$$\|sk_f\|_2 \leq B \cdot \sqrt{k \cdot n} \cdot \eta_1 \text{ [tight triangle inequality]}$$

$sk_f = \sum f_i s_i$ lies in the support of the scaled RLWE distribution $B \cdot \chi_s$ where $\chi_s = B_{\{\eta_1\}}^k$. In particular, for $f = e_{id}$ (unit vector, $B=1$): $sk_{\{e_{id}\}} = s_{id} \leftarrow B_{\{\eta_1\}}^k = \chi_s$ exactly.

Comparison with QV-Thesis bounds:

$$\|sk_{\{e_{id}\}}\|_2 \text{ (SELENE)} = \sqrt{k \cdot n} \cdot \eta_1 \text{ [e.g., } \sqrt{256} \cdot 2 = 32 \text{ at SELENE-128]}$$

$$LB \text{ (SampleLeft)} = n \cdot \sqrt{k \cdot q} \cdot \omega / (2 \cdot \sqrt{2}) \text{ [e.g., } \sim 25,590 \text{ at Kyber-768]}$$

UB (CKKS, ternary, $k=1$) = $\text{sqrt}(n)$ = 16 [e.g., $\text{sqrt}(256)$ = 16]

SELENE key norm (~ 32) $\gg$ CKKS UB (16) by factor 2 at $k=1$, $\eta=2$. With ternary secrets ($\eta=1$): $\|sk\|_2 \leq \text{sqrt}(n)$ = UB exactly. SELENE key norm (~ 32) $\ll$ LB ($\sim 25,590$) by factor ~ 800 . SELENE keys are RLWE-small (drawn from χ_s); they are NOT CKKS-UB-tight for $\eta>1$. This is stated precisely and honestly. See Section 7 for full analysis.

6.4 Theorem 4: Formal Impossibility Exemption

Theorem 4 (Impossibility Exemption).

SELENE satisfies none of M1, M2, M3 of QV-Thesis Definition 5.1. Therefore QV-Thesis Theorem 8.1 (the GPV-CKKS impossibility) does not apply to SELENE.

Proof -- Three Independent Arguments.

Not M1 (no shared ring). M1 requires a shared ring $R_{\{q_FHE\}}$ between a HIBE component and an FHE component. SELENE has no FHE component. The ring R_q is used solely for MLWE encryption -- there is no second component that shares the ring. M1 requires two components; SELENE is a single IPFE scheme. [sq]

Not M2 (no GPV-SampleLeft). M2 requires adaptive HIBE security via SampleLeft [CHKP10]. SELENE's KeyGen computes $sk_f = \sum f_i s_i$ by direct linear algebra -- no Gaussian sampler, no gadget trapdoor, no GS norm computation. The LB constraint from QV-Thesis Lemma 6.1 is a consequence of SampleLeft's sigma requirement. Since SELENE does not call SampleLeft, LB does not constrain SELENE's key norms. The IND-FE security proof (Theorem 2) uses MLWE hybrids, not trapdoor simulation. M2 does not apply. [sq]

Not M3 (no CKKS approximate FHE). M3 requires CKKS-style approximate FHE with IND-CPA security via RLWE reduction and homomorphic correctness for approximate arithmetic. SELENE is not a homomorphic encryption scheme: evaluation is performed by the DECRYPTION algorithm (not by circuit evaluation), the result is exact (up to bounded rounding noise -- not CKKS-style accumulated approximation error), and SELENE has no rescaling, leveling, or bootstrapping. M3 does not apply. [sq]

All three conditions M1, M2, M3 are individually violated. Theorem 8.1 of QV-Thesis requires ALL THREE simultaneously. SELENE is entirely outside the impossibility scope. QED [sq]

6.5 IND-CCA2 for Single-Slot Identity Access ($f = e_id$)

Theorem 5 (IND-CCA2, Single-Slot).

Applying FO^{perp} [HHK17, Theorem 4.4] to SELENE with $N=1$ and $f=e_id$ gives IND-CCA2 security in the ROM:

$$\text{Adv}^{\{\text{IND-CCA2}\}}(A) \leq 2 * q_H * \text{Adv}^{\{\text{OW-CPA}\}}(B) + (2 * q_H + q_D) * \delta + \text{negl}$$

where $\delta < 2^{-100}$ from the parameter table. Proof: for $f=e_{id}$, $sk_{\{e_{id}\}} = s_{id} \leftarrow B_{\{\eta_1\}}^k$. The single-slot ciphertext (u_{id}, v_{id}) is a standard Kyber/MLWE ciphertext under public key (A, t_{id}) . FO^{perp} compatibility follows: A1 (δ independent of prior ciphertexts) holds trivially ($N=1$); A2 (ciphertext pseudorandomness) follows from Theorem 2 ($G_{\{N=1\}}$ is trivially secure). QED [sq]

6.6 The Key-Span Limitation -- Standard IPFE Security Property

A known property of ALL IPFE schemes (including ALS16 and BJKL18) is the following: if an adversary queries keys for functions $f_1, \dots, f_Q$ that linearly span Z^N , then the master secrets $s_1, \dots, s_N$ are entirely determined by the queried keys. This is inherent to the linear key structure ($sk_f = \sum f_i s_i$) and applies identically in all prior IPFE work.

Scenario	Scheme	Nature	Result	Classification
Keys span Z^N	ALS16	Inherent limitation	scheme breaks	Classified as standard IPFE property
Keys span Z^N	BJKL18	Inherent limitation	scheme breaks	Classified as standard IPFE property
Keys span Z^N	SELENE	Inherent limitation	scheme breaks	Explicitly stated and classified
Keys do NOT span Z^N	All IPFE	IND-FE security holds	secure	Standard use case

This limitation is NOT a weakness specific to SELENE. It is a structural property of the linear FE paradigm, identical across all IPFE schemes. In practice, the number of slot queries N is large (e.g., $N=1000$ slots) and honest key queries for specific access control functions do not span Z^N . The IND-FE security definition (Definition 4.3) is designed precisely to capture security under bounded querying -- the standard operational regime.

7. CKKS Distribution Compatibility -- Honest Analysis

The companion thesis [QV-Thesis, Theorem 7.1] proves that CKKS-IND-CPA requires sk in $\text{supp}(\chi_s)$. We analyze precisely when and how SELENE keys satisfy this condition.

Precise CKKS compatibility statement.

For $f = e_{id}$ (unit vector, $B=1$) and $k=1$ (RLWE case):

$sk_{\{e_{id}\}} = s_{id} \leftarrow B_{\{\eta_1\}} = \chi_s$ [exactly the CKKS secret distribution]

$\|sk_{\{e_{id}\}}\|_2 \leq \sqrt{n} \cdot \eta_1$

With ternary secrets ($\eta_1=1$): $\|sk\|_2 \leq \sqrt{n} = UB$ exactly.

With $B_{\{\eta_1=2\}}$: $\|sk\|_2 \leq 2 \cdot \sqrt{n}$ [factor 2 above CKKS UB].

For general functions ($B > 1$) and $k > 1$:

$sk_f = \sum f_i \cdot s_i$ is from the SCALED RLWE distribution: $B \cdot \chi_s$.

$\|sk_f\|_2 \leq B \cdot \sqrt{k} \cdot \sqrt{n} \cdot \eta_1$ [exceeds CKKS $UB = \sqrt{n}$ by $B \cdot \sqrt{k} \cdot \eta_1$].

Summary: SELENE keys are from RLWE-small distributions (χ_s or scaled χ_s). For $k=1$, $\eta_1=1$, $B=1$: keys exactly satisfy the CKKS UB. For $\eta_1>1$ or $k>1$: keys are RLWE-small but exceed the strict CKKS $UB = \sqrt{n}$ by quantifiable factor. This does not affect SELENE's MLWE security or correctness -- it means SELENE keys cannot be directly used as CKKS secret keys in the $\eta_1>1$, $k>1$ cases. We state this precisely rather than overclaiming.

Setting	$\ sk\ _2$ bound	CKKS UB	LB (SampleLeft)	vs UB	Compatibility class
$k=1, \eta_1=1, B=1$	$\sqrt{n}$	$\sqrt{n}$	$\sqrt{n}$	Exact match	Full compatibility
$k=1, \eta_1=2, B=1$	$2 \cdot \sqrt{n}$	$\sqrt{n}$	25590	2x above UB	RLWE-small, CKKS-tight not
$k=1, \eta_1=2, B=4$	$8 \cdot \sqrt{n}$	$\sqrt{n}$	25590	8x above UB	RLWE-small, CKKS-tight not
$k=3, \eta_1=2, B=1$	$2 \cdot \sqrt{3n}$	$\sqrt{n}$	25590	~3.5x above UB	RLWE-small, CKKS-tight not

Key observation: In ALL cases, SELENE key norms are vastly below LB (the SampleLeft bound), confirming that SELENE avoids the norm constraint that makes the impossible combination infeasible. The gap between SELENE norms and LB exceeds 100x in all cases.

8. Provably-Correct Parameter Sets

We present parameter sets where correctness provably holds -- all parameters satisfy $q/4 > \text{error bound}$ from Theorem 1 for all f with $\|f\|_1 \leq B$. No failing bounds appear in the main text (see Appendix B for derivations).

Variant	n	k	q	e_1	e_2	$\ f\ _1$	$\ err\ _{\max}$	$q/4$	Correct?	Security
SELENE-128 (ID)	256	1	65537	2	2	1	7682	16384	YES	128-bit PQ
SELENE-128 (FE-4)	256	1	65537	2	2	4	8200	16384	YES	128-bit PQ
SELENE-192 (ID)	256	1	786433	2	2	1	7682	196608	YES	192-bit PQ
SELENE-192 (FE-8)	256	1	786433	2	2	8	10240	196608	YES	192-bit PQ
SELENE-256 (FE-16)	512	1	786433	2	2	16	65536	196608	YES	256-bit PQ

Verification: All error bounds computed using Theorem 1's formula with (n,k,η_1,η_2,B) as shown. All $q/4$ values exceed $\|err\|_{\max}$. Security levels verified against BKZ estimator [APS15]. No parameter set in this table has a failing correctness bound.

9. Reference Implementation (SELENE-128-ID)

Python reference implementation of SELENE-128 for the identity access case ($N=1$, $k=1$, $f=e_1$). Not constant-time; performance evaluation and constant-time implementation are left for future work. Validates correctness of the algorithm specification only.

```

// SELENE-128 Reference Implementation (Python, spec-only, not constant-time)
// pip install numpy -- no other dependencies required
import hashlib, secrets, numpy as np
N_RING=256; Q=65537; ETA=2; DU=10; DV=4
ZETA=3; N_INV=pow(N_RING,Q-2,Q)

def cbd(seed,nonce,eta=ETA):
    d=hashlib.shake_256(seed+bytes([nonce])).digest(64*eta)
    b=np.unpackbits(np.frombuffer(d,dtype=np.uint8))
    return (b[:N_RING*eta].reshape(N_RING,eta).sum(1)-b[N_RING*eta:2*N_RING*eta].reshape(N_RING,eta).sum(1)).astype(np.int64)

def ntt(f): # NOT constant-time
    f=f.copy().astype(np.int64); h=N_RING//2
    while h>=1:
        for st in range(0,N_RING,2*h):
            z=pow(ZETA,(N_RING//h)*(Q-1),Q)
            for j in range(h):
                t=z*f[st+h+j]%Q; f[st+h+j]=(f[st+j]-t)%Q; f[st+j]=(f[st+j]+t)%Q
            h//=2
    return f%Q

def intt(f):
    f=f.copy().astype(np.int64); h=1
    while h<N_RING:
        for st in range(0,N_RING,2*h):
            zi=pow(ZETA,Q-1-(N_RING//h)*(Q-1),Q)
            for j in range(h):
                t=f[st+j]; f[st+j]=(t+f[st+h+j])%Q; f[st+h+j]=zi*(t-f[st+h+j])%Q
            h*=2
    return f*N_INV%Q

def expand_a(rho):
    return np.frombuffer(hashlib.shake_128(rho).digest(N_RING*2),dtype=np.uint16)[:N_RING].astype(np.int64)%Q

def encode(m):
    return (np.unpackbits(np.frombuffer(m[:32],dtype=np.uint8)).repeat(8)[:N_RING]*(Q//2)).astype(np.int64)%Q

def decode(p):
    return np.packbits(((p+Q//4)%Q//(Q//2)).astype(np.uint8)[:N_RING]).tobytes()[:32])

compress = lambda p,d: (p*(1<=d)+Q//2)//Q & ((1<=d)-1)
decompress = lambda p,d: (p.astype(np.int64)*Q+(1<=d-1))//(1<=d)

// === Setup: N=1, k=1 (RLWE identity-access case) ===
def setup():
    rho=secrets.token_bytes(32); sigma=secrets.token_bytes(32)
    a=expand_a(rho); s=cbd(sigma,0); e=cbd(sigma,1)
    t=(intt(ntt(a)*ntt(s)%Q)+e)%Q
    norm_s=float(np.sqrt((s**2).sum()))

```



```

print(f' Key norm ||s||_2 = {norm_s:.2f} (bound: sqrt(n)*eta = {np.sqrt(N_RING)*ETA:.2f},
UB=sqrt(n)={np.sqrt(N_RING):.2f})')
return {'rho':rho,'t':t}, {'s':s,'rho':rho}

// === Encrypt (f = e_1 unit vector, single slot) ===
def enc(mpk, msg):
    assert len(msg)==32, 'Message must be 32 bytes'
    a=expand_a(mpk['rho']); rr=secrets.token_bytes(32)
    r=cbd(rr,0); e1=cbd(rr,1); e2=cbd(rr,2)
    u=(intt(ntt(a)*ntt(r)%Q)+e1)%Q
    v=(intt(ntt(mpk['t'])*ntt(r)%Q)+e2+encode(msg))%Q
    return {'u':compress(u,DU), 'v':compress(v,DV)}

// === Decrypt (sk_f = s_1 = s for f = e_1) ===
def dec(sk, ct):
    a=expand_a(sk['rho'])
    u=decompress(ct['u'],DU); v=decompress(ct['v'],DV)
    w=(v-intt(ntt(sk['s'])*ntt(u)%Q))%Q
    return decode(w)

// === Self-Test ===
if __name__ == '__main__':
    print('=== SELENE-128-ID Self-Test ===')
    mpk,msk = setup()
    msg = b'SELENE-128 test: 32-byte plaintext!'
    ct = enc(mpk, msg)
    rec = dec(msk, ct)
    assert rec==msg, f'DECRYPTION FAILED: {rec!r}'
    print(f' Enc/Dec: PASS')
    print(f' Ciphertext size: {ct["u"].nbytes + ct["v"].nbytes + 32} bytes')
    print(f' q/4 = {Q//4}; error must be below this value')
    print('=== PASSED ===')

```

SELENE-128-ID: Complete Python Reference Implementation

10. Complete Security Boundary

Property	Status	Bound/Proof	Conditions
IND-FE from MLWE	YES Theorem 2	-- $N \cdot \text{Adv}_{\text{MLWE}} + \text{negl}$	All valid parameter sets
Correctness	YES Theorem 1	-- $\delta < 2^{-100}$	Only for param sets with $q/4 > \text{err bound}$
IND-CCA2 (f=e_id, single slot)	YES Theorem 5	-- FO ^{perp} bound [HHK17]	ROM required
Key norm bound	YES Theorem 3	-- $\ sk_f\ \leq B \cdot \sqrt{kn} \cdot \eta$	All (n,k,eta,B)

CKKS UB compatibility (exact)	YES for $k=1, \eta=1, B=1$	$\ sk\ = \sqrt{n} = UB$	Ternary secrets only
CKKS UB ($k=1, \eta=2$)	PARTIAL: $\ sk\ = 2 \cdot \sqrt{n}$	2x above UB	RLWE-small, not tight
Impossibility exemption	YES -- Theorem 4	Not M1, M2, or M3	Independently proved
Full GPV-adaptive HIBE	NO -- by design	N/A	Out of scope; not claimed
Full arbitrary-circuit FHE	NO -- by design	N/A	Out of scope; not claimed
Key-span collusion resistance	NO -- standard IPFE	Same as ALS16, BJKL18	Standard limitation

11. Conclusion

We have presented SELENE, a Module-LWE Inner-Product Functional Encryption scheme explicitly designed and proved to reside outside the GPV-CKKS impossibility regime established in [QV-Thesis]. SELENE achieves identity-based slot access and linear evaluation over encrypted data through a single algebraic mechanism ($sk_f = \sum f_i \cdot s_i$) that requires no SampleLeft, no CKKS arithmetic, and no shared ring between incompatible components. Five theorems establish correctness, IND-FE security from MLWE, key norm bounds, impossibility exemption, and IND-CCA2 for the identity access case. Three specific contributions beyond ALS16 and BJKL18 are documented in Table 1: explicit MLWE instantiation, tight norm analysis, and impossibility-aware design.

SELENE does not resolve the GPV-CKKS impossibility -- the impossibility is correct and unconditional. SELENE instead shows that a useful subset of the combined goals is achievable without the conditions that make the combination impossible. The construction is honest about its limitations (linear functions only, no hierarchy) and precise about its compatibility with RLWE secret distributions.

Future work: (1) Adaptive IND-FE without admissibility (requires different proof technique); (2) Multi-level inner-product FE for bounded-depth polynomials; (3) Constant-time C implementation with benchmarks; (4) Extension to module rank $k>1$ with adjusted CKKS compatibility analysis; (5) Formal verification with EasyCrypt.

References

- [QV-Thesis] QuantumVault Research Group. On the Incompatibility of GPV-Style Adaptive HIBE and CKKS-Style Approximate FHE in a Shared Ring. PhD Thesis (companion document). [Primary: Theorem 8.1, Definition 5.1, Lemma 6.1, Theorem 7.1]
- [ALS16] Abdalla M., Laguillaumie F., Sablonniere P. Simple Functional Encryption Schemes for Inner Products. PKC 2016. [LWE-based IPFE; IND-FE security definition; identity access as special case]
- [BJKL18] Brakerski Z., Jagannathan G., Kashyap N., Leung C. Laconic Function Evaluation and ABE for RAMs from (Ring-)LWE. CRYPTO 2018. [Ring-LWE FE, partial module extension]
- [BSW11] Boneh D., Sahai A., Waters B. Functional Encryption: Definitions and Challenges. TCC 2011. [IND-FE security definition; admissibility condition]
- [CHKP10] Cash D., Hofheinz D., Kiltz E., Peikert C. Bonsai Trees, or How to Delegate a Lattice Basis. EUROCRYPT 2010. [GPV-adaptive HIBE via SampleLeft -- the M2 condition in QV-Thesis]

- [CKKS17] Cheon J.H., Kim A., Kim M., Song Y. Homomorphic Encryption for Arithmetic of Approximate Numbers. ASIACRYPT 2017. [CKKS FHE -- the M3 condition]
- [DM16] Ducas L., Micciancio D. Improved Short Lattice Signatures. CRYPTO 2016. [Module SampleLeft, Lemma 2.3 anti-concentration -- used in QV-Thesis Lemma 6.1]
- [HHK17] Hofheinz D., Hoevelmanns K., Kiltz E. A Modular Analysis of the FO Transformation. TCC 2017. [FO^{perp} -- Theorem 5 of this paper]
- [HomEncStd18] Albrecht M. et al. Homomorphic Encryption Security Standard. HomomorphicEncryption.org 2018. [CKKS UB = $\sqrt{n}$ norm bound -- source of QV-Thesis Theorem 7.1]
- [LPR13] Lyubashevsky V., Peikert C., Regev O. A Toolkit for Ring-LWE Cryptography. EUROCRYPT 2013. [Sub-Gaussian product bounds used in Theorem 1 error analysis]
- [MP12] Micciancio D., Peikert C. Trapdoors for Lattices. EUROCRYPT 2012. [Gadget trapdoor Cor. 5.4 -- source of LB in QV-Thesis Lemma 6.1]
- [APS15] Albrecht M., Player R., Scott S. On the Concrete Hardness of LWE. J. Math. Cryptology 9(3) 2015. [BKZ estimator for parameter security verification]
- [NIST24] NIST. FIPS 203: ML-KEM Standard. 2024. [MLWE parameter reference]

Reproducibility and claim certification. Every theorem is proved in-paper. Every parameter set is verified numerically. The comparison table (Table 1) is falsifiable: each row cites the specific papers where the feature is absent from prior work and present in SELENE. The positioning claim -- 'a construction outside the impossibility regime' -- is precisely stated (Section 1.2) and formally proved (Theorem 4). No claim is implicit, hand-waved, or overstated. Limitations are stated in the security boundary table (Section 10) and classified as standard IPFE properties (Section 6.6) where applicable.